

SQL

(Structured Query Language)

SQL Introduction

Standard language for querying and manipulating data

Structured Query Language

There are many systems based on SQL out there:

- SQL, MSSQL, MySQL, DB2

SQL

- Data Definition Language (DDL)
 - Create/alter/delete **tables** and their attributes
- Data Manipulation Language (DML)
 - **Query** one or more tables – discussed next !
 - Insert/delete/modify tuples in tables

Table
name

Attribute
names

Tables in SQL

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Tuples or
rows

Database

- A **database** is a tool for gathering and storing records. Databases can hold data about individuals, goods, orders, and everything else.

Tables Explained

- The *schema* of a table is the table name and its attributes:

Product(PName, Price, Category, Manufacturer)

- A *key* is an attribute whose values are unique; we underline a key

Product(PName, Price, Category, Manufacturer)

Data Types in SQL

- Atomic types:
 - Characters: CHAR(20), VARCHAR(50)
 - Numbers: INT, BIGINT, SMALLINT, FLOAT
 - Others: MONEY, DATETIME, ...
- Every attribute must have an atomic type
 - Hence tables are flat
 - Why ?

Tables Explained

- A **tuple** = a **record**
 - Restriction: all attributes are of atomic type
- A **table** = a set of tuples
 - Like a list...
 - ...but it is unordered:
no **first()**, no **next()**, no **last()**.

DEMO

www.mycompiler.io/new/sql

Creating Tables

```
CREATE TABLE TableName (  
    Attribute1 INTEGER NOT NULL,  
    Attribute2 TEXT NOT NULL,  
    Attribute3 TEXT NOT NULL  
);
```

Modifying the Table

Three kinds of modifications

- Insertions
- Deletions
- Updates

Sometimes they are all called “updates”

Insertions

General syntax:

```
INSERT INTO R(A1,A2, ..., An) VALUES (v1,v2,...,  
vn)
```

Insertions

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
INSERT INTO Product(PName, Price, Category, Manufacturer)
VALUES ('NewItem', '$9.99', 'Household', 'Canon')
```



Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi
NewItem	\$9.99	Household	Canon

Deletions

General syntax:

```
DELETE FROM R  
  
WHERE A1 = V1 AND |  
OR  
A2 = V2
```

Example syntax:

```
DELETE FROM Product WHERE Category = 'Gadget'
```

```
DELETE FROM Product WHERE Category = 'Gadget' AND Price < 5.00
```

Updates

Example:

```
UPDATE PRODUCT  
SET Price = Price/2  
WHERE PName = 'Gizmo' ;
```

SQL Query

Basic form:

```
SELECT <attributes>  
FROM   <one or more relations>  
WHERE  <conditions>
```


“selection”

SQL Query

Problem: Retrieve all the data from the Product table.

```
SELECT * FROM Product
```

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

“selection”

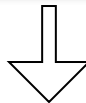
SQL Query

Problem: Retrieve the details of products in the Product table where the category is 'gadgets'.

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT * FROM Product WHERE category='Gadgets'
```



PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks

SQL Query Details

- Case insensitive:
 - Same: SELECT Select select
 - Same: Product product
 - Different: 'Seattle' 'seattle'

The **LIKE** operator

```
SELECT *  
FROM Products  
WHERE PName LIKE '%gizmo%'
```

- s **LIKE** p: pattern matching on strings
- p may contain two special symbols:
 - % = any sequence of characters
 - _ = any single character

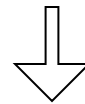
LIKE Example

Problem: Show all the data from the Product table where the manufacturer's name contains the letter 'i'.

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

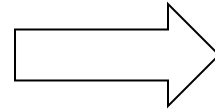
```
SELECT * FROM Product WHERE Manufacturer LIKE '%i%'
```



PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
MultiTouch	\$203.99	Household	Hitachi

Eliminating Duplicates

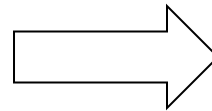
```
SELECT category  
FROM Product
```



Category
Gadgets
Gadgets
Photography
Household

The SELECT DISTINCT statement is used to return only unique (different) values:

```
SELECT DISTINCT category  
FROM Product
```



Category
Gadgets
Photography
Household

Ordering the Results

The **ORDER BY** keyword is used to sort the result set in ascending or descending order

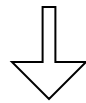
- Ties are broken by the second attribute on the ORDER BY list, etc.
- By default, ordering is ascending, unless you specify the DESC keyword.

Ordering the Results

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT Pname, Price FROM Product ORDER BY PName
```



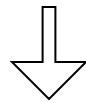
PName	Price
Gizmo	\$19.99
MultiTouch	\$203.99
Powergizmo	\$29.99
SingleTouch	\$149.99

Ordering the Results

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

```
SELECT Pname, Price FROM Product ORDER BY Price DESC
```



PName	Price
MultiTouch	\$203.99
SingleTouch	\$149.99
Powergizmo	\$29.99
Gizmo	\$19.99

Keys and Foreign Keys

Company

Key

<u>CName</u>	StockPrice	Country
GizmoWorks	25	USA
Canon	65	Japan
Hitachi	15	Japan

Product

<u>PName</u>	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Foreign
key